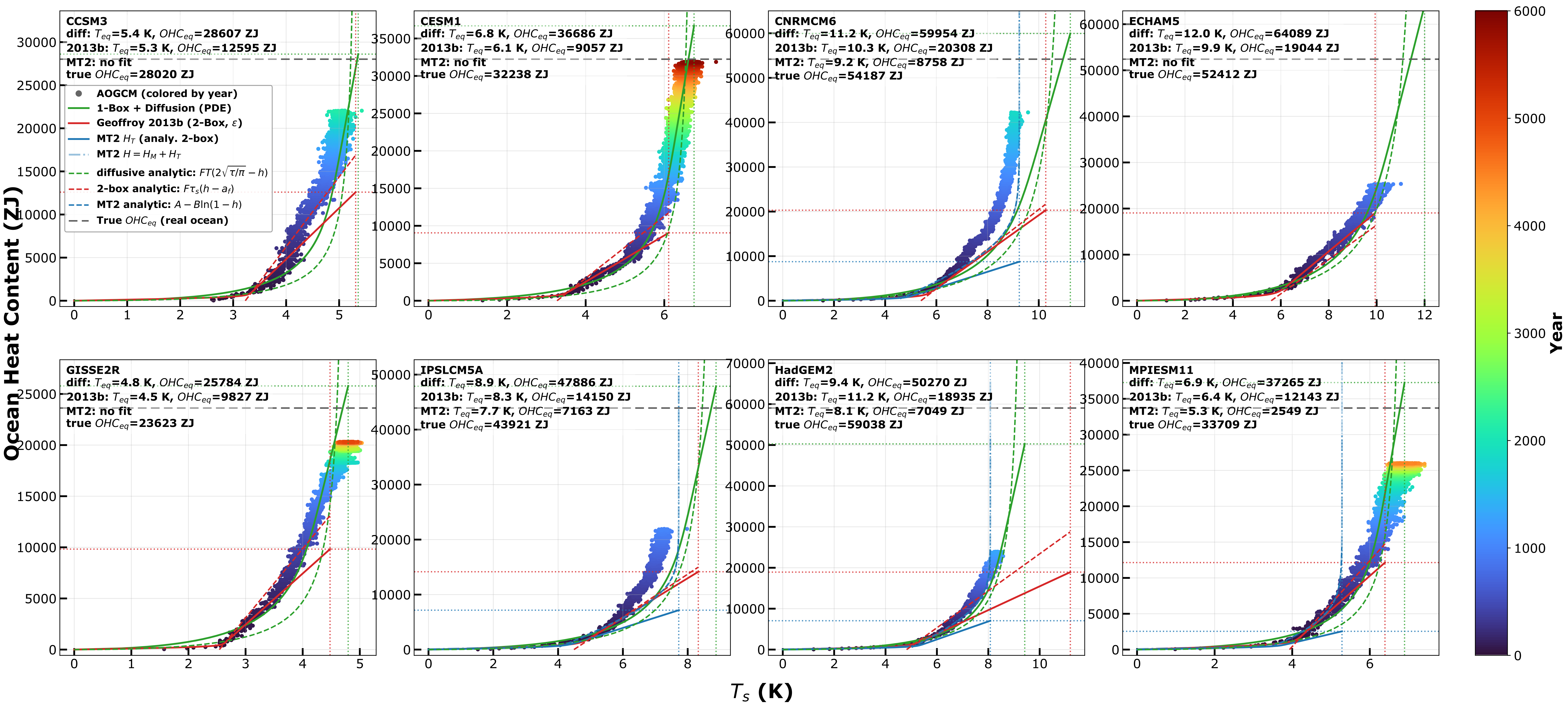
4xCO$_2$ OHC vs. Surface Temp: AOGCMs and three fitted models

01_replicate parameters


CCSM3
diff: T_{eq}=5.4 K, OHC_{eq}=28607 ZJ
2013b: T_{eq}=5.3 K, OHC_{eq}=12595 ZJ
MT2: no fit
true OHC_{eq}=28020 ZJ
AOGCM (colored by year)
1-Box + Diffusion (PDE)
Geoffroy 2013b (2-Box, ϵ)
MT2 H_T (analy. 2-box)
MT2 $H = H_M + H_T$
diffusive analytic: $FT(2\sqrt{\tau/\pi} - h)$
2-box analytic: $F\tau_s(h - a_f)$
MT2 analytic: $A - B\ln(1 - h)$
True OHC_{eq} (real ocean)
CESM1
diff: T_{eq}=6.8 K, OHC_{eq}=36686 ZJ
2013b: T_{eq}=6.1 K, OHC_{eq}=9057 ZJ
MT2: no fit
true OHC_{eq}=32238 ZJ
CNRMCM6
diff: T_{eq}=11.2 K, OHC_{eq}=59954 ZJ
2013b: T_{eq}=10.3 K, OHC_{eq}=20308 ZJ
MT2: T_{eq}=9.2 K, OHC_{eq}=8758 ZJ
true OHC_{eq}=54187 ZJ
ECHAM5
diff: T_{eq}=12.0 K, OHC_{eq}=64089 ZJ
2013b: T_{eq}=9.9 K, OHC_{eq}=19044 ZJ
MT2: no fit
true OHC_{eq}=52412 ZJ
GISSE2R
diff: T_{eq}=4.8 K, OHC_{eq}=25784 ZJ
2013b: T_{eq}=4.5 K, OHC_{eq}=9827 ZJ
MT2: no fit
true OHC_{eq}=23623 ZJ
IPSLCM5A
diff: T_{eq}=8.9 K, OHC_{eq}=47886 ZJ
2013b: T_{eq}=8.3 K, OHC_{eq}=14150 ZJ
MT2: T_{eq}=7.7 K, OHC_{eq}=7163 ZJ
true OHC_{eq}=43921 ZJ
HadGEM2
diff: T_{eq}=9.4 K, OHC_{eq}=50270 ZJ
2013b: T_{eq}=11.2 K, OHC_{eq}=18935 ZJ
MT2: T_{eq}=8.1 K, OHC_{eq}=7049 ZJ
true OHC_{eq}=59038 ZJ
MPIESM11
diff: T_{eq}=6.9 K, OHC_{eq}=37265 ZJ
2013b: T_{eq}=6.4 K, OHC_{eq}=12143 ZJ
MT2: T_{eq}=5.3 K, OHC_{eq}=2549 ZJ
true OHC_{eq}=33709 ZJ
Ocean Heat Content (ZJ)
T_s (K)
Year
0
1000
2000
3000
4000
5000
6000